

Classification of UCI Wine Quality (Red + White) - Decision Tree

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- **Subject Name:** Machine Learning
- **Subject Code:** MLN 601
- **Title:** Binary Classification of Wine Quality - Decision Tree vs Naive Bayes
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- **Assessment No.:** 2
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Field	Value
Dataset	UCI Wine Quality - red + white combined (Cortez et al., 2009)
Task	Binary classification: low (quality < 6) vs high (quality >= 6)
Algorithm	Decision Tree (tuned, required) + Logistic Regression and Gaussian Naive Bayes comparators
Tuning	GridSearchCV, 5-fold cross-validation, scoring = ROC AUC
Evaluation	AUC-ROC curve, confusion matrix, precision / recall / F1
Methodology	CRISP-DM

This notebook follows the six CRISP-DM stages to classify a wine as **high** or **low** quality from its physicochemical measurements. Assessment 1 modelled the quality score as regression; this brief revisits the same data as **classification**, tuning a Decision Tree judged primarily on the **AUC-ROC** curve. This version also adds a generative **Gaussian Naive Bayes** model (Module 5) as a Bayesian contrast to the discriminative tree.

Quality is a subjective, ordinal score, so the classifier is framed as an **interpretable decision-support tool** that flags likely low-quality batches for human review, not a replacement for expert tasting.

1. Business Understanding

Wine producers and distributors want a fast, low-cost screen that flags batches likely to score poorly with tasters, so limited expert time is spent where it matters. The **business question** for this assessment is therefore: *can a wine be classified as high or low quality from routine physicochemical lab measurements alone?*

The UCI Wine Quality dataset records 11 physicochemical traits (acidity, residual sugar, alcohol, and so on) and an expert **quality** score from 0-10 for red and white *vinho verde* wines (Cortez et al., 2009). Following the brief, the score is collapsed into a **binary target: below 6 is "low" (class 1), 6 or above is "high" (class 0)**. Low is the *positive* class because flagging weak batches is the action of interest.

Success is defined relative to a naive baseline that always predicts the majority class. A useful model must beat that floor and, more importantly, achieve a high **AUC-ROC** - the chance it ranks a random low-quality wine above a random high-quality one - since AUC is threshold-independent and robust to class imbalance. The output supports, but does not replace, human quality control.

2. Data Understanding

This stage acquires the data, checks its quality, and explores how the physicochemical features relate to wine quality. The red and white files are loaded from UCI (or a local copy) and concatenated into one frame with a `wine_type` indicator (1 = red, 0 = white) - about 6,500 wines for the classifier.

```
In [1]: import os
import warnings

# Silence sklearn/joblib parallel UserWarnings so they cannot flood the expc
# PDF. n_jobs=1 in the grid search below means no worker processes spawn eit
os.environ["PYTHONWARNINGS"] = "ignore"
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.linear_model import LogisticRegression
from sklearn.dummy import DummyClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, roc_auc_score, roc_curve,
                             confusion_matrix, ConfusionMatrixDisplay,
                             classification_report)

sns.set_theme(style="whitegrid")
RANDOM_STATE = 42
```

```
In [2]: from pathlib import Path

# Resolve project folders relative to the notebook location so the same note
```

```

# runs locally and on Colab/Azure without editing paths.
NB_DIR = Path.cwd()
BASE_DIR = NB_DIR.parent if NB_DIR.name == "notebook" else NB_DIR
OUTPUT_DIR = BASE_DIR / "outputs"
FIG_DIR = OUTPUT_DIR / "figures"
FIG_DIR.mkdir(parents=True, exist_ok=True)

UCI_BASE = ("https://archive.ics.uci.edu/ml/machine-learning-databases/"
            "wine-quality/")

def load_wine(colour):
    fname = f"winequality-{colour}.csv"
    candidates = [BASE_DIR / "dataset" / fname,
                  NB_DIR / "dataset" / fname,
                  NB_DIR / fname]
    local = next((p for p in candidates if p.exists()), None)
    if local is not None:
        print("Loading local:", local)
        return pd.read_csv(local, sep=";")
    print("Loading from UCI URL:", UCI_BASE + fname)
    return pd.read_csv(UCI_BASE + fname, sep=";")

red = load_wine("red")
white = load_wine("white")
red["wine_type"] = 1 # 1 = red
white["wine_type"] = 0 # 0 = white
df = pd.concat([red, white], ignore_index=True)

print("Red:", red.shape, "| White:", white.shape, "| Combined:", df.shape)
df.head()

```

Loading local: /Users/luisfaria/Desktop/sEngineer/masters_SWEAI/2026-T2/MLN/assignments/Assessment2/dataset/winequality-red.csv
Loading local: /Users/luisfaria/Desktop/sEngineer/masters_SWEAI/2026-T2/MLN/assignments/Assessment2/dataset/winequality-white.csv
Red: (1599, 13) | White: (4898, 13) | Combined: (6497, 13)

Out [2]:

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphat
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.

In [3]: df.info()

```
<class 'pandas.DataFrame'>
RangeIndex: 6497 entries, 0 to 6496
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   fixed acidity                         6497 non-null   float64
1   volatile acidity                     6497 non-null   float64
2   citric acid                          6497 non-null   float64
3   residual sugar                       6497 non-null   float64
4   chlorides                           6497 non-null   float64
5   free sulfur dioxide                 6497 non-null   float64
6   total sulfur dioxide                6497 non-null   float64
7   density                             6497 non-null   float64
8   pH                                  6497 non-null   float64
9   sulphates                          6497 non-null   float64
10  alcohol                             6497 non-null   float64
11  quality                             6497 non-null   int64
12  wine_type                           6497 non-null   int64
dtypes: float64(11), int64(2)
memory usage: 660.0 KB
```

```
In [4]: df.describe().T
```

	count	mean	std	min	25%	50%	75%
fixed acidity	6497.0	7.215307	1.296434	3.80000	6.40000	7.00000	7.70000
volatile acidity	6497.0	0.339666	0.164636	0.08000	0.23000	0.29000	0.40000
citric acid	6497.0	0.318633	0.145318	0.00000	0.25000	0.31000	0.39000
residual sugar	6497.0	5.443235	4.757804	0.60000	1.80000	3.00000	8.10000
chlorides	6497.0	0.056034	0.035034	0.00900	0.03800	0.04700	0.06500
free sulfur dioxide	6497.0	30.525319	17.749400	1.00000	17.00000	29.00000	41.00000
total sulfur dioxide	6497.0	115.744574	56.521855	6.00000	77.00000	118.00000	156.00000
density	6497.0	0.994697	0.002999	0.98711	0.99234	0.99489	0.99699
pH	6497.0	3.218501	0.160787	2.72000	3.11000	3.21000	3.32000
sulphates	6497.0	0.531268	0.148806	0.22000	0.43000	0.51000	0.60000
alcohol	6497.0	10.491801	1.192712	8.00000	9.50000	10.30000	11.30000
quality	6497.0	5.818378	0.873255	3.00000	5.00000	6.00000	6.00000
wine_type	6497.0	0.246114	0.430779	0.00000	0.00000	0.00000	0.00000

```
In [5]: print("Missing values per column:")
print(df.isna().sum().to_string())
print("\nDuplicate rows:", int(df.duplicated().sum()))
```

```
Missing values per column:
fixed acidity          0
volatile acidity       0
citric acid            0
residual sugar         0
chlorides              0
free sulfur dioxide    0
total sulfur dioxide    0
density               0
pH                    0
sulphates             0
alcohol               0
quality               0
wine_type             0
```

```
Duplicate rows: 1177
```

2.2 The classification target

The brief defines the target by thresholding the expert score: **quality < 6 -> "low" (1)** and **quality >= 6 -> "high" (0)**. Encoding low quality as the positive class (1) keeps the model's "positive" prediction aligned with the business action - flagging a batch for review. The cell below derives the `quality_label` column and shows the class balance. The split is uneven (more high-quality wines), which is exactly why **accuracy alone is misleading** and the evaluation leans on AUC-ROC and per-class recall.

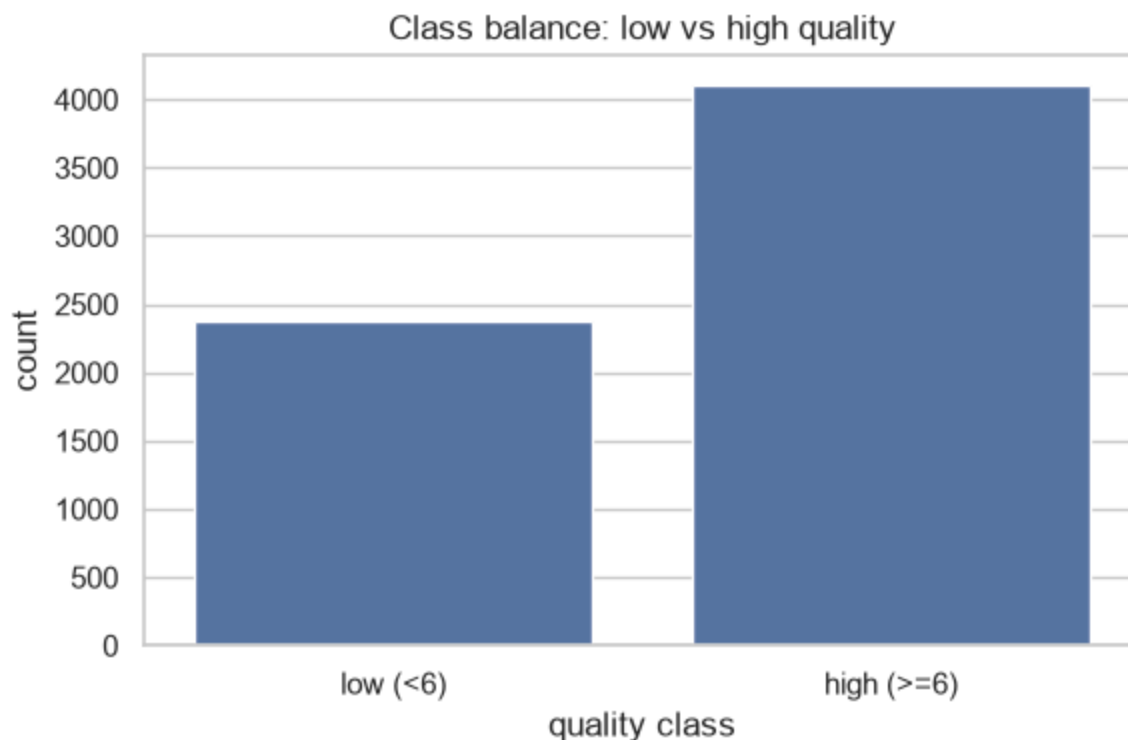
```
In [6]: # Brief's encoding: quality < 6 -> low (1, positive class); quality >= 6 ->
df["quality_label"] = (df["quality"] < 6).astype(int)
label_names = {0: "high (>=6)", 1: "low (<6)"}
df["quality_name"] = df["quality_label"].map(label_names)

counts = df["quality_label"].value_counts().sort_index()
print("Class counts:")
print(counts.rename(index=label_names).to_string())
print("\nClass proportions:")
print((counts / len(df)).round(3).rename(index=label_names).to_string())

plt.figure(figsize=(6, 4))
sns.countplot(x="quality_name", data=df, order=["low (<6)", "high (>=6)"])
plt.title("Class balance: low vs high quality")
plt.xlabel("quality class")
plt.ylabel("count")
plt.tight_layout()
plt.savefig(FIG_DIR / "v2_class_balance.png", dpi=120)
plt.show()
```

```
Class counts:
quality_label
high (>=6)    4113
low (<6)       2384
```

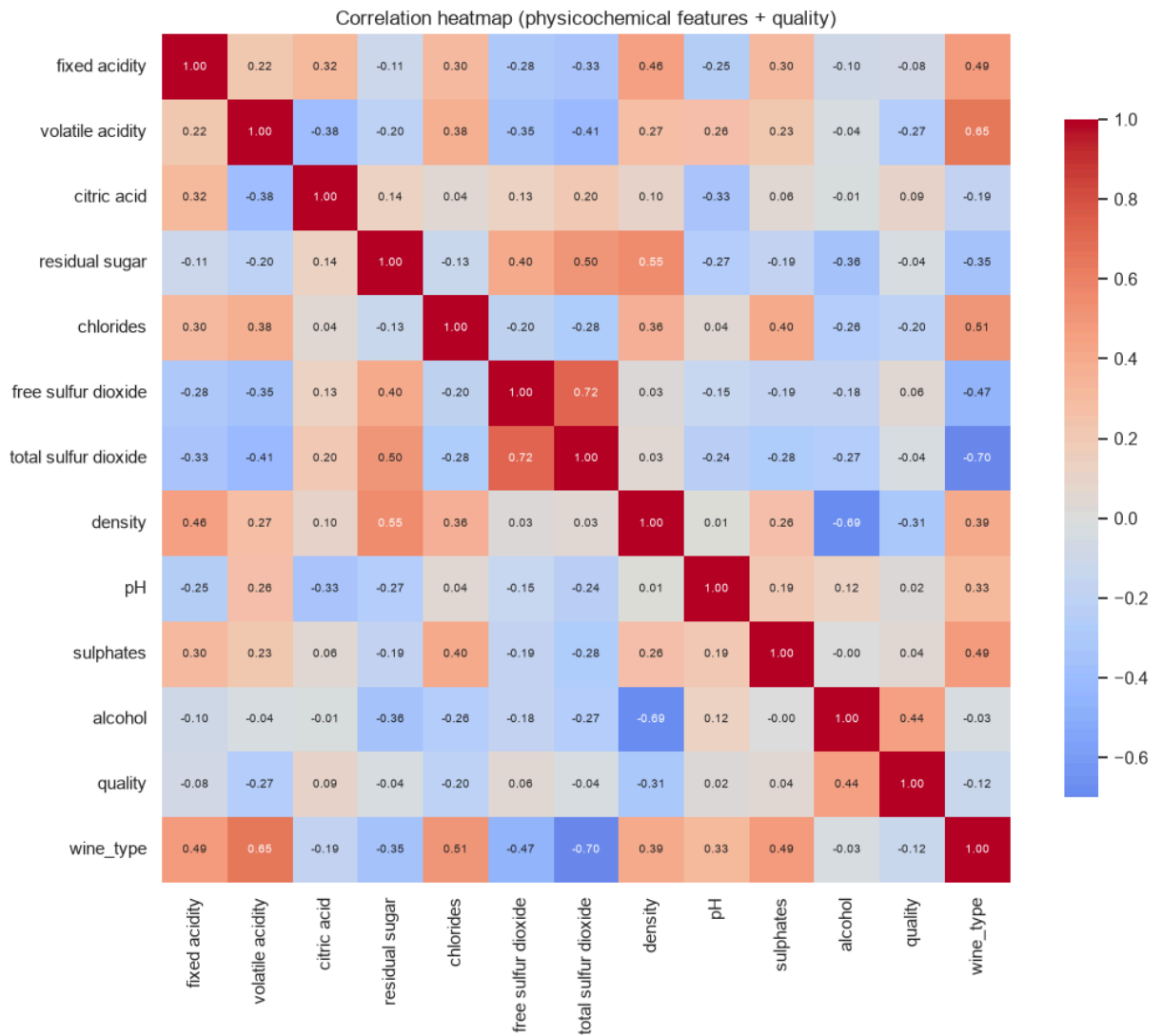
```
Class proportions:
quality_label
high (>=6)     0.633
low (<6)       0.367
```



2.3 Feature relationships

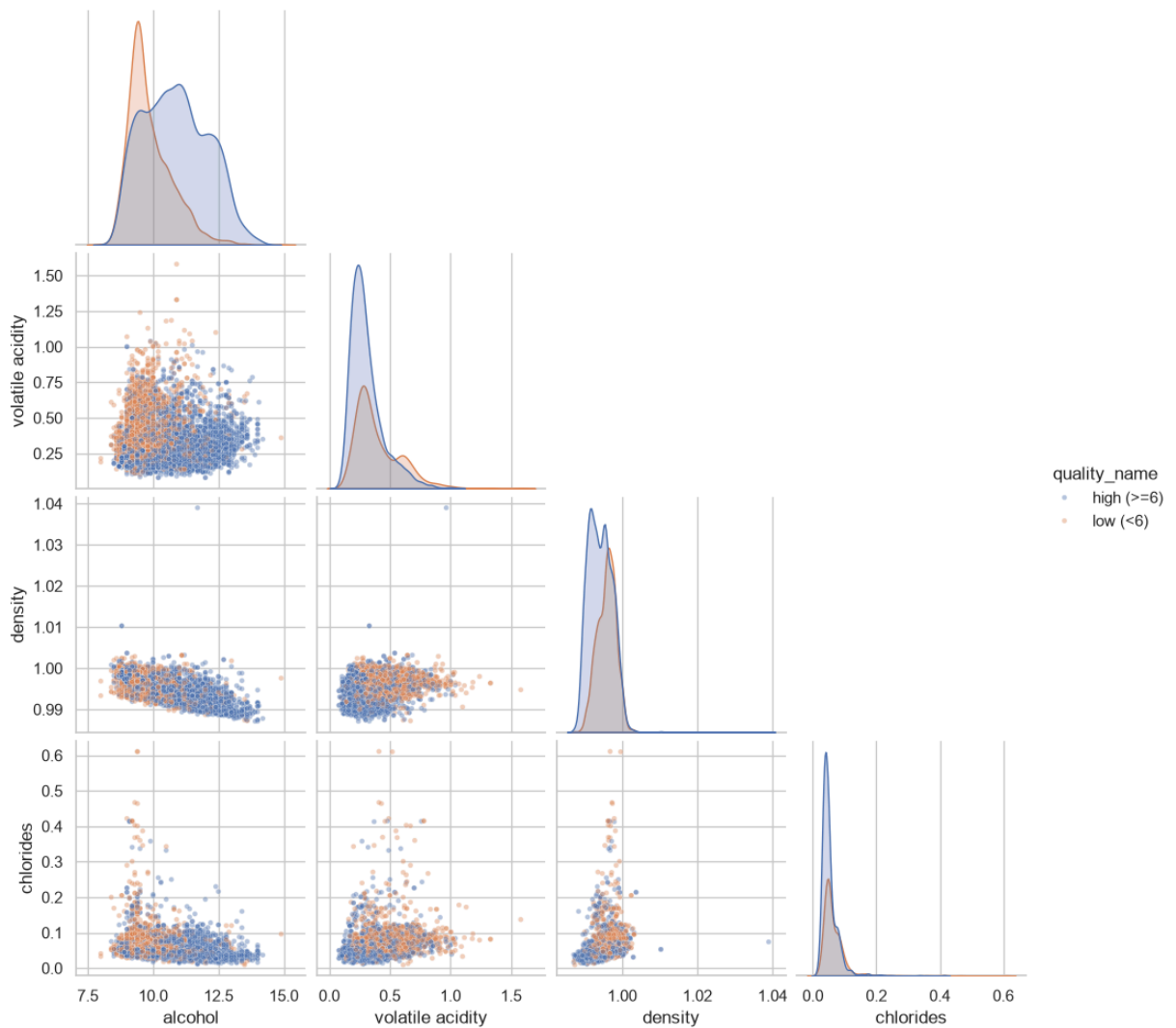
The correlation heatmap and the seaborn **pairplot** (required by the brief) show how the features move with each other and with quality. `alcohol` shows the clearest separation between high and low wines, while several acidity and sulfur-dioxide measures are correlated with each other - a hint that a non-linear model able to capture interactions may do well.

```
In [7]: plt.figure(figsize=(11, 9))
corr = df.drop(columns=["quality_label"]).corr(numeric_only=True)
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", center=0,
            square=True, cbar_kws={"shrink": 0.8}, annot_kws={"size": 7})
plt.title("Correlation heatmap (physicochemical features + quality)")
plt.tight_layout()
plt.savefig(FIG_DIR / "v2_corr_heatmap.png", dpi=120)
plt.show()
```



```
In [8]: pair_cols = ["alcohol", "volatile acidity", "density", "chlorides"]
g = sns.pairplot(df, vars=pair_cols, hue="quality_name",
                 hue_order=["high (>=6)", "low (<6)"], corner=True,
                 plot_kws={"alpha": 0.4, "s": 12}, diag_kind="kde")
g.fig.suptitle("Pairplot of key features by quality class", y=1.02)
g.savefig(FIG_DIR / "v2_pairplot.png", dpi=110)
plt.show()
```

Pairplot of key features by quality class



Interpretation:

Three things stand out:

- First, there are **no missing values**, so no imputation is needed.
- Second, `alcohol` has the strongest positive association with high quality and `volatile acidity` the clearest negative one, matching wine-science intuition.
- Third, several features are **mutually correlated** (for example `free sulfur dioxide` with `total sulfur dioxide`, and `density` with `alcohol` and `residual sugar`).

Strong collinearity destabilises a linear model's coefficients but barely affects a Decision Tree, which splits one feature at a time on thresholds. The pairplot also shows the two classes **overlap heavily** - no single feature separates them cleanly - so we should expect a good-but-imperfect classifier rather than a near-perfect one.

3. Data Preparation

Preparation here is light because the data is already clean and numeric. Three steps matter. **(1) Feature/target split:** `quality` and the derived label columns are removed from the feature matrix `X` so the original score cannot leak into training; `wine_type` is kept as an informative feature. **(2) Train/test split:** an 80/20 **stratified** split preserves the high/low ratio in both partitions, which matters under class imbalance, with a fixed `random_state` for reproducibility. **(3) Scaling:** a Decision Tree is **scale-invariant** (it compares each feature against a threshold), so it needs no scaling; the Logistic Regression comparator *does*, so its `StandardScaler` is placed **inside a Pipeline**. That way the scaler is refit on each training fold during cross-validation and never sees the validation rows - preventing the data leakage that would otherwise inflate the scores. The earlier check flagged **1,177 duplicate rows**; with no unique wine ID these may be distinct wines with identical readings, so they are kept to avoid discarding valid samples.

```
In [9]: LEAK_COLS = ["quality", "quality_label", "quality_name"]
feature_cols = [c for c in df.columns if c not in LEAK_COLS]
X = df[feature_cols]
y = df["quality_label"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=RANDOM_STATE, stratify=y)

print("Train:", X_train.shape, "| Test:", X_test.shape)
print("\nTrain class balance:")
print(y_train.value_counts(normalize=True).round(3).rename(index=label_names))
print("Test class balance:")
print(y_test.value_counts(normalize=True).round(3).rename(index=label_names))
print("\nFeatures used (%d):" % len(feature_cols), feature_cols)
```

```
Train: (5197, 12) | Test: (1300, 12)
```

```
Train class balance:
```

```
quality_label
```

```
high (>=6)    0.633
```

```
low (<6)      0.367
```

```
Test class balance:
```

```
quality_label
```

```
high (>=6)    0.633
```

```
low (<6)      0.367
```

```
Features used (12): ['fixed acidity', 'volatile acidity', 'citric acid', 'residual sugar', 'chlorides', 'free sulfur dioxide', 'total sulfur dioxide', 'density', 'pH', 'sulphates', 'alcohol', 'wine_type']
```

4. Modelling

Five models are compared so the tuned Decision Tree can be read against both a *discriminative* and a *generative* alternative:

1. **Majority-class baseline** (`DummyClassifier`) - the floor every real model must beat; its AUC sits at ~0.5 by construction.

2. **Default Decision Tree** - an untuned tree, to show what tuning actually buys.
3. **Tuned Decision Tree** - the required model. A `GridSearchCV` searches `max_depth`, `min_samples_leaf`, `criterion` (gini vs entropy) and `class_weight` (none vs balanced) using **5-fold cross-validation**, scored on **ROC AUC**. An unconstrained tree grows until its leaves are pure and overfits; these hyperparameters prune that complexity.
4. **Logistic Regression** - a linear probabilistic comparator (Module 8). The brief asks for one technique, so this is included only as **optional context**: a straight-line decision boundary to contrast with the tree's axis-aligned splits.
5. **Gaussian Naive Bayes** - a *generative* comparator (Module 5's Bayes rule). Where the others learn the boundary directly, it models each feature's distribution per class and inverts it with Bayes' theorem - a different philosophy on the same data.

Every estimator uses a fixed seed and `n_jobs=1`. Cross-validation tunes on the training set only; the test set is touched once, in Evaluation.

```
In [10]: # 1) Majority-class baseline
dummy = DummyClassifier(strategy="most_frequent")
dummy.fit(X_train, y_train)

# 2) Default (untuned) Decision Tree
dt_default = DecisionTreeClassifier(random_state=RANDOM_STATE)
dt_default.fit(X_train, y_train)

# 3) Tuned Decision Tree - the required model
param_grid = {
    "max_depth": [3, 4, 5, 6, 8, None],
    "min_samples_leaf": [1, 5, 10, 20],
    "criterion": ["gini", "entropy"],
    "class_weight": [None, "balanced"],
}
grid = GridSearchCV(
    DecisionTreeClassifier(random_state=RANDOM_STATE),
    param_grid, scoring="roc_auc", cv=5, n_jobs=1)
grid.fit(X_train, y_train)
dt_best = grid.best_estimator_
print("Best params:", grid.best_params_)
print("Best CV ROC AUC: %.3f" % grid.best_score_)

# 4) Logistic Regression comparator (needs scaling -> Pipeline)
logreg = Pipeline([
    ("scaler", StandardScaler()),
    ("clf", LogisticRegression(max_iter=2000, random_state=RANDOM_STATE)),
])
logreg.fit(X_train, y_train)

# 5) Gaussian Naive Bayes - generative comparator (Module 5, Bayes rule)
nb = GaussianNB()
nb.fit(X_train, y_train)

models = {
```

```

    "Majority baseline": dummy,
    "Decision Tree (default)": dt_default,
    "Decision Tree (tuned)": dt_best,
    "Logistic Regression": logreg,
    "Naive Bayes (Gaussian)": nb,
}
print("\nFitted models:", list(models))

```

Best params: {'class_weight': None, 'criterion': 'gini', 'max_depth': 6, 'min_samples_leaf': 20}
 Best CV ROC AUC: 0.796

Fitted models: ['Majority baseline', 'Decision Tree (default)', 'Decision Tree (tuned)', 'Logistic Regression', 'Naive Bayes (Gaussian)']

4.2 Selected hyperparameters

`GridSearchCV` refits the best configuration on the full training set. The chosen parameters balance depth against generalisation - a moderate `max_depth` with a leaf size large enough to stop the tree memorising individual wines. Saved to `outputs/` for traceability.

```

In [11]: best_params = pd.Series(grid.best_params_, dtype="object").astype(str)
best_params.loc["cv_roc_auc"] = round(grid.best_score_, 4)
best_params.to_csv(OUTPUT_DIR / "tree_best_params_v2.csv", header=["value"])
best_params.to_frame("value")

```

```

Out[11]:

```

	value
class_weight	NaN
criterion	gini
max_depth	6
min_samples_leaf	20
cv_roc_auc	0.7959

5. Evaluation

The models are now judged on the **held-out test set** they never saw during tuning. Because the classes are imbalanced, no single number tells the whole story, so several complementary metrics are reported:

- **AUC-ROC** - the headline metric: how well the model *ranks* low-quality wines above high-quality ones across all thresholds, which makes it robust to imbalance.
- **Accuracy** - intuitive but misleading here: always predicting "high" already scores 63%, so it must be read against the baseline.
- **Precision, recall and F1 for the low class** - whether the model actually *catches* the weak batches, and how often its flags are correct.

- **Confusion matrix** - the raw counts behind those rates.

The ROC curves are overlaid (diagonal = random guessing); the tree plot and feature-importance chart keep the model's logic transparent - a white-box advantage.

```
In [12]: def class_metrics(model, X_te, y_te):
    pred = model.predict(X_te)
    if hasattr(model, "predict_proba"):
        proba = model.predict_proba(X_te)[:, 1]  # P(class 1 = low)
    else:
        proba = model.decision_function(X_te)
    pos = dict(pos_label=1, zero_division=0)
    return {
        "accuracy": accuracy_score(y_te, pred),
        "precision_low": precision_score(y_te, pred, **pos),
        "recall_low": recall_score(y_te, pred, **pos),
        "f1_low": f1_score(y_te, pred, **pos),
        "roc_auc": roc_auc_score(y_te, proba),
    }

results = pd.DataFrame(
    {name: class_metrics(m, X_test, y_test) for name, m in models.items()}
).T.round(3)
results.index.name = "model"
results.to_csv(OUTPUT_DIR / "model_metrics_v2.csv")
results
```

Out[12]:

	accuracy	precision_low	recall_low	f1_low	roc_auc
model					

model	accuracy	precision_low	recall_low	f1_low	roc_auc
Majority baseline	0.633	0.000	0.000	0.000	0.500
Decision Tree (default)	0.786	0.702	0.725	0.713	0.773
Decision Tree (tuned)	0.737	0.637	0.656	0.647	0.809
Logistic Regression	0.752	0.700	0.568	0.627	0.814
Naive Bayes (Gaussian)	0.703	0.615	0.509	0.557	0.771

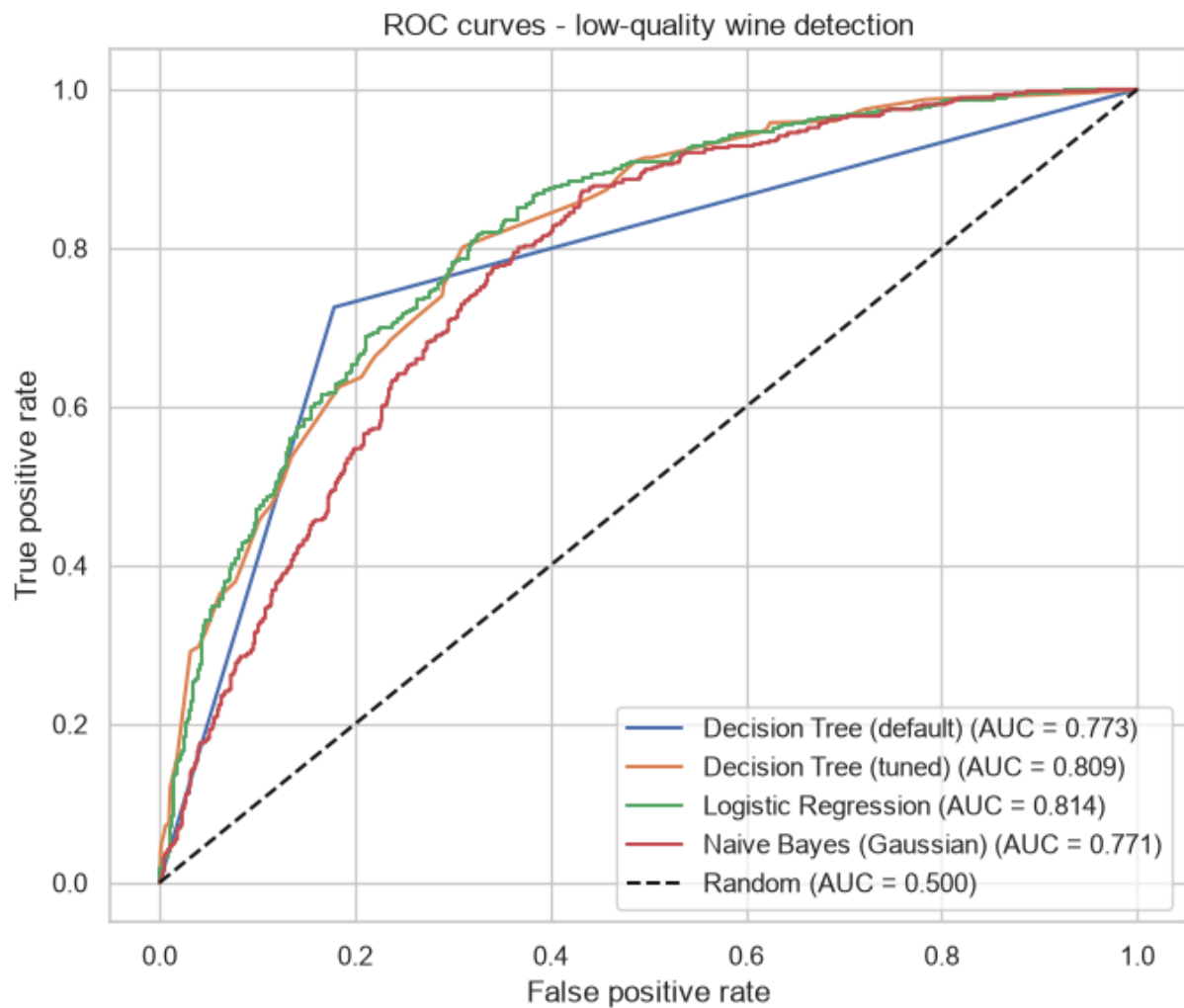
```
In [13]: report = classification_report(
    y_test, dt_best.predict(X_test),
    target_names=["high (>=6)", "low (<6)"],
    output_dict=True, zero_division=0)
report_df = pd.DataFrame(report).T.round(3)
report_df.to_csv(OUTPUT_DIR / "classification_report_v2.csv")
report_df
```

Out [13]:

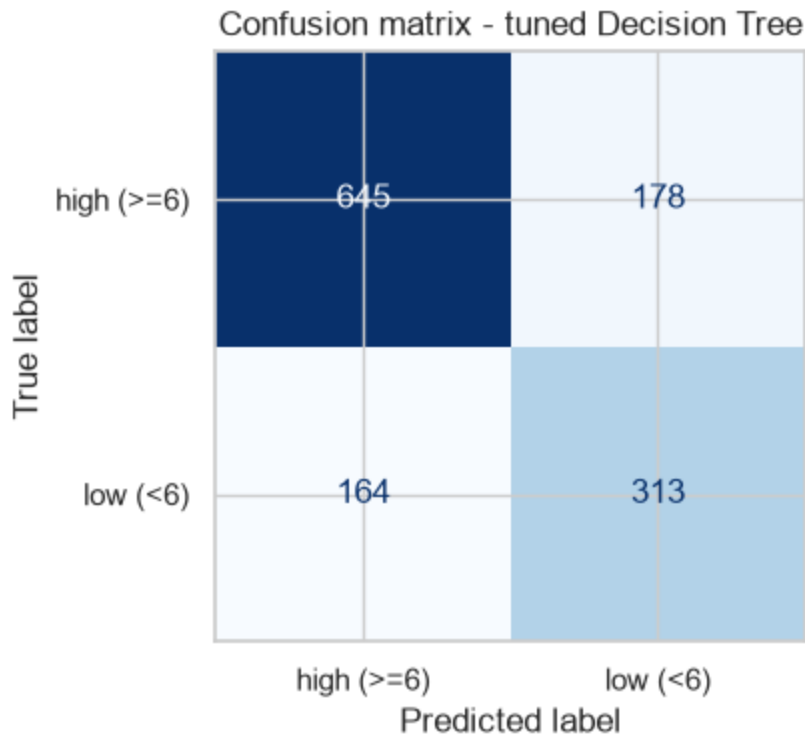
	precision	recall	f1-score	support
high (>=6)	0.797	0.784	0.790	823.000
low (<6)	0.637	0.656	0.647	477.000
accuracy	0.737	0.737	0.737	0.737
macro avg	0.717	0.720	0.719	1300.000
weighted avg	0.739	0.737	0.738	1300.000

In [14]:

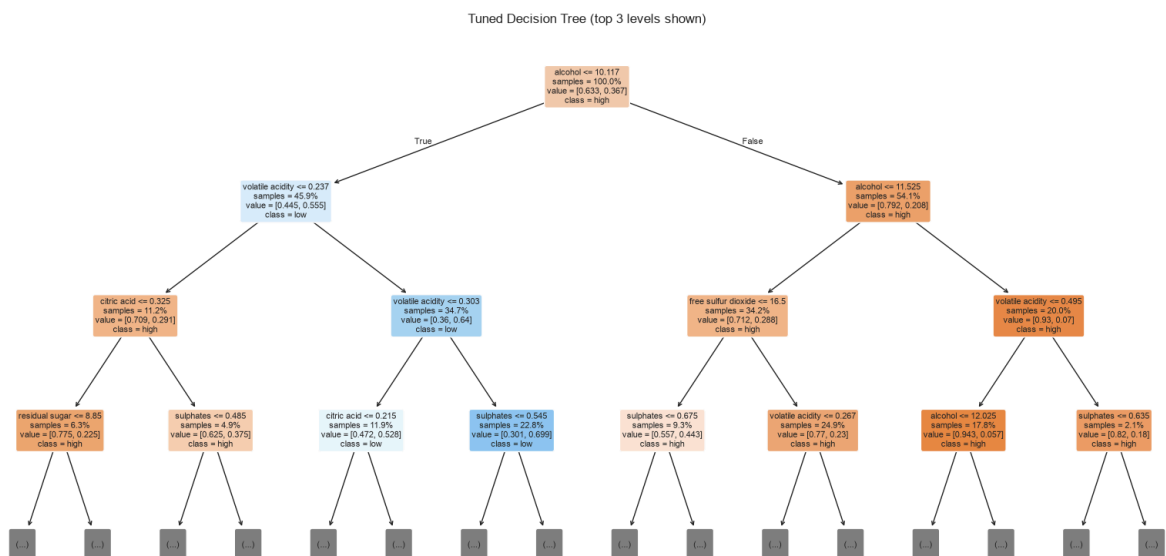
```
plt.figure(figsize=(7, 6))
for name, m in models.items():
    if name == "Majority baseline":
        continue
    proba = m.predict_proba(X_test)[: , 1]
    fpr, tpr, _ = roc_curve(y_test, proba, pos_label=1)
    auc = roc_auc_score(y_test, proba)
    plt.plot(fpr, tpr, label=f"{name} (AUC = {auc:.3f})")
plt.plot([0, 1], [0, 1], "k--", label="Random (AUC = 0.500)")
plt.xlabel("False positive rate")
plt.ylabel("True positive rate")
plt.title("ROC curves - low-quality wine detection")
plt.legend(loc="lower right")
plt.tight_layout()
plt.savefig(FIG_DIR / "v2_roc_curves.png", dpi=120)
plt.show()
```



```
In [15]: fig, ax = plt.subplots(figsize=(5, 4))
cm = confusion_matrix(y_test, dt_best.predict(X_test))
ConfusionMatrixDisplay(cm, display_labels=["high (>=6)", "low (<6)"]).plot(
    ax=ax, cmap="Blues", colorbar=False)
ax.set_title("Confusion matrix - tuned Decision Tree")
plt.tight_layout()
plt.savefig(FIG_DIR / "v2_confusion_matrix.png", dpi=120)
plt.show()
```

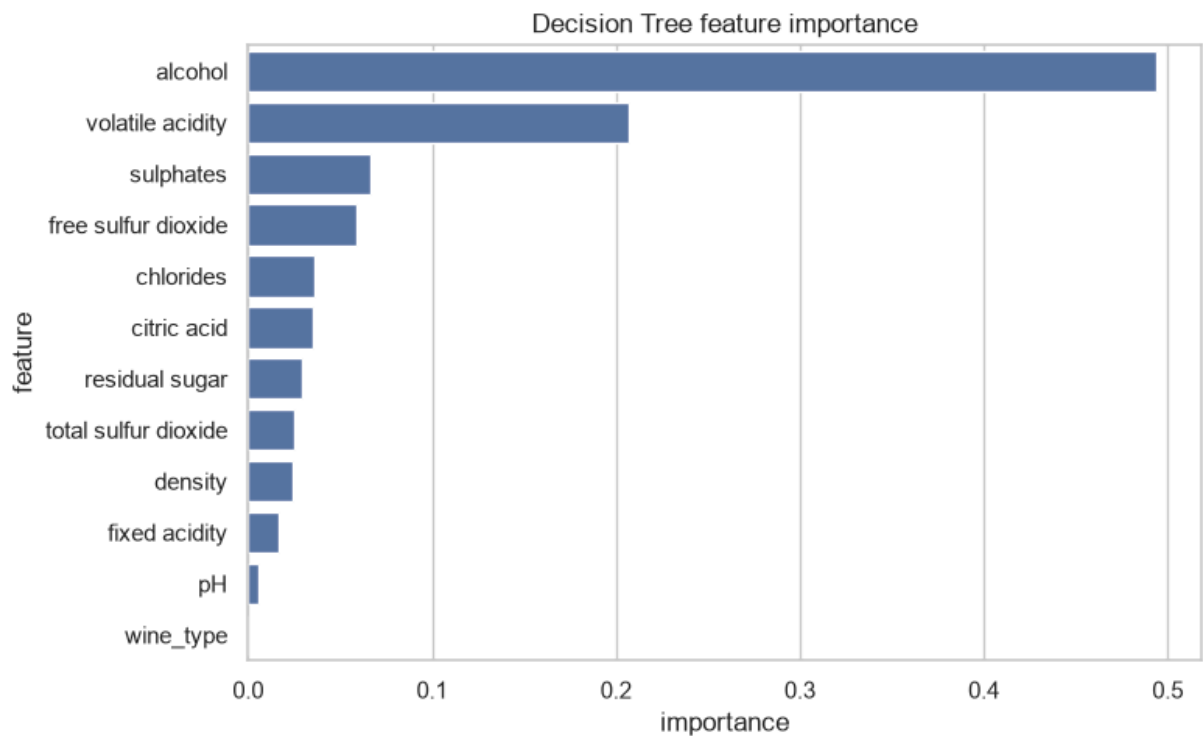


```
In [16]: plt.figure(figsize=(16, 8))
plot_tree(dt_best, feature_names=feature_cols, class_names=["high", "low"],
          filled=True, rounded=True, max_depth=3, fontsize=8,
          impurity=False, proportion=True)
plt.title("Tuned Decision Tree (top 3 levels shown)")
plt.tight_layout()
plt.savefig(FIG_DIR / "v2_decision_tree.png", dpi=110)
plt.show()
```



```
In [17]: importance = (pd.DataFrame({"feature": feature_cols,
                                   "importance": dt_best.feature_importances_})
               .sort_values("importance", ascending=False)
               .reset_index(drop=True))
importance.to_csv(OUTPUT_DIR / "feature_importance_v2.csv", index=False)
```

```
plt.figure(figsize=(8, 5))
sns.barplot(x="importance", y="feature", data=importance)
plt.title("Decision Tree feature importance")
plt.tight_layout()
plt.savefig(FIG_DIR / "v2_feature_importance.png", dpi=120)
plt.show()
importance
```



Out[17]:

	feature	importance
0	alcohol	0.493657
1	volatile acidity	0.207295
2	sulphates	0.066892
3	free sulfur dioxide	0.059251
4	chlorides	0.036338
5	citric acid	0.035003
6	residual sugar	0.029429
7	total sulfur dioxide	0.025249
8	density	0.024245
9	fixed acidity	0.016697
10	pH	0.005944
11	wine_type	0.000000

Interpretation:

The tuned Decision Tree reaches a test **AUC of 0.809**, far above the majority baseline (0.500) and essentially level with the Logistic Regression comparator (0.814). Tuning raised the tree's AUC over the untuned default (0.773) by pruning it back (`max_depth=6` , `min_samples_leaf=20`) from a fully grown, overfit tree - though it trades a little raw accuracy (0.74 vs 0.79) for that better, more honest ranking. The confusion matrix shows the model confirms high-quality wines (recall 0.78) more reliably than it catches every low-quality one (recall 0.66) - the residual cost of class imbalance and heavy class overlap. **Feature importance** is dominated by `alcohol` (0.49) and `volatile acidity` (0.21), echoing the EDA and matching wine chemistry, while `wine_type` adds almost nothing. With an AUC near 0.81 the model is a useful screen but far from perfect: it ranks wines well, yet should not auto-reject a batch on its own.

5.4 Generative vs discriminative: where Naive Bayes lands

The Decision Tree and Logistic Regression are **discriminative** - they model the boundary $P(\text{class} \mid \text{features})$ directly. **Gaussian Naive Bayes** is **generative**: it learns how each feature is distributed per class and inverts it with Bayes' rule (Module 5). Its core assumption - features *independent given the class* - is **violated here**, since the EDA showed strong correlations (the two sulfur-dioxide measures at 0.72). Yet it still reaches an AUC of **0.771**, the lowest of the real models but still well clear of random and level with the untuned tree (0.773). That echoes the classic result that Naive Bayes is surprisingly robust for *ranking* even when independence breaks, because the decision needs the right order, not calibrated probabilities (Domingos & Pazzani, 1997).

6. Deployment / Lessons Learned

In CRISP-DM, deployment for this assessment means reflection rather than a production release.

What worked: framing the problem as binary classification made the Decision Tree a natural, interpretable fit, and AUC-ROC gave an honest read of performance under imbalance, where accuracy would have flattered a lazy "always high" model. The white-box tree and feature-importance chart make the result explainable to a non-technical stakeholder - a real advantage for quality control.

What was harder: the classes overlap heavily in feature space, so no amount of tuning produced a clean separation; the model is a *screen*, not an oracle. The imbalance shows up as weaker recall on the minority "low" class (0.66 vs 0.78); the grid search explored `class_weight="balanced"` to counter it, but ultimately favoured a shallower, pruned tree that maximised AUC.

What I would do differently next time: test an ensemble (Random Forest or gradient boosting) to see how much a single tree leaves on the table; tune the decision threshold

explicitly to the cost of missing a bad batch rather than defaulting to 0.5; and engineer interaction features the tree currently has to discover on its own. The honest lesson echoes Assessment 1 - **model and framing choices, plus careful evaluation, matter more than raw data volume.**

Academic Integrity Declaration

I declare that except where I have referenced, the work I am submitting for this assessment task is my own work. I have read and am aware of the Torrens University Australia Academic Integrity Policy and Procedure viewable online at <http://www.torrens.edu.au/policies-and-forms>

I am aware that I need to keep a copy of all submitted material and their drafts, and I will do so accordingly.

Statement of Acknowledgement

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- Anthropic Claude Opus 4.8
- OpenAI ChatGPT Codex 5.5

Both tools were used to assist with understanding ML concepts, structuring the technical pipeline, improving clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1. "Explain how to convert the UCI wine quality score into a binary classification target and why AUC-ROC is the right metric when the classes are imbalanced."
2. "How do I tune a scikit-learn DecisionTreeClassifier with GridSearchCV (max_depth, min_samples_leaf, criterion, class_weight) using 5-fold cross-validation scored on ROC AUC, without leaking the test set?"
3. "Why is a Decision Tree scale-invariant while Logistic Regression needs a StandardScaler, and how should I place the scaler in a Pipeline to avoid data leakage during cross-validation?"

I confirm that the use of these tools has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

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